
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print):

Jesse Russell

Student ID:

2240

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	6
2	10	2
3	10	2
4	10	2
5	10	5
Total:	50	17

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

2

$$\iiint_T f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw$$

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

2

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dA = \oint_C \vec{F} \cdot d\vec{r}$$

OK, I'll accept

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

1

$$\vec{N} = \nabla(z - x^2 - y^2) = \begin{bmatrix} -2x \\ -2y \\ 1 \end{bmatrix}$$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

0

$$\vec{F} = (x^2 + y^2) \hat{i} - 2y \hat{j}$$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

1

$$r^2 + z^2 = x^2 + y^2 + z^2$$

(calculus)

Why?

X

$$\therefore (f \cos \phi)^2 + (f \sin \phi \sin \phi)^2 + (f \cos \phi)^2$$

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Cylindrical $h_1 = 1$
 $h_2 = r$
 $h_3 = 1$
 Jacobian = r

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial x} \left(\frac{r}{1} \frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{1}{r} \frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{r}{1} \frac{\partial f}{\partial z} \right) \right]$$

$$= \frac{2r}{r} + \frac{2}{r^2} + \frac{2r}{r}$$

$$= 2 + \frac{2}{r^2} + 2$$

Use formula from
 formula sheet?

Spherical $h_1 = 1$
 $h_2 = r \sin \phi$
 $h_3 = r$
 Jacobian = $r^2 \sin \phi$

$$\nabla^2 f = \frac{1}{r^2 \sin \phi} \left[\frac{\partial}{\partial r} \left(\frac{r^2 \sin \phi}{1} \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \phi} \left(\frac{1}{r \sin \phi} \frac{\partial f}{\partial \phi} \right) + \frac{\partial}{\partial \theta} \left(\frac{r \sin \phi}{r} \frac{\partial f}{\partial \theta} \right) \right]$$

$$= \frac{1}{r^2 \sin \phi} \left[2r^2 \sin \phi + \frac{2}{\sin \phi} + 2 \sin \phi \right]$$

$$= \frac{2r^2 \sin \phi}{r^2 \sin \phi} + \frac{2}{r^2 \sin^2 \phi} + \frac{2 \sin \phi}{r^2 \sin \phi}$$

$$= 2 + \frac{2}{r^2 \sin^2 \phi} + \frac{2}{r^2}$$

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.

$$\phi = \text{const}$$

$$dA = \rho^2 \sin \phi \, d\phi \, d\theta \, d\rho$$

$$= \frac{1}{2} \rho^2 \, d\phi \, d\theta \, d\rho \quad \rho \sin \phi \, d\theta \, d\rho$$

$$\iint \vec{F} \cdot \vec{n} \, dA$$

$$\phi = \frac{\pi}{6}$$

$$0 \leq \rho \leq 2$$

$$\int_0^{2\pi} \int_0^{2\pi} \int_0^2$$

$$d\phi \, d\theta \, d\rho \quad \vec{F} = L \quad ? \quad X$$

$$2\pi$$

□

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .*Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus*

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

$$F = [x, y, z]$$

$$\nabla \cdot F = \frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z = 3$$

$$= [1, 1, 1] \times$$

$$\vec{F} = [\rho \cos \theta \sin \phi, \rho \sin \theta \sin \phi, \rho \cos \theta]$$

$$\int_0^{2\pi} \int_0^\pi \int_0^2 \rho \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{2\pi} \int_0^\pi \frac{\rho^2}{2} \Big|_0^2 \, d\theta \, d\phi$$

$$= 2 \int_0^{2\pi} \int_0^\pi \rho \, d\theta \, d\phi = 4\pi \rho$$

$$\vec{h} = \vec{r}_\theta \times \vec{r}_\phi$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial \theta} x & \frac{\partial}{\partial \theta} y & \frac{\partial}{\partial \theta} z \\ \frac{\partial}{\partial \phi} x & \frac{\partial}{\partial \phi} y & \frac{\partial}{\partial \phi} z \end{vmatrix} = 0$$

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Continue problem 4

Solution.

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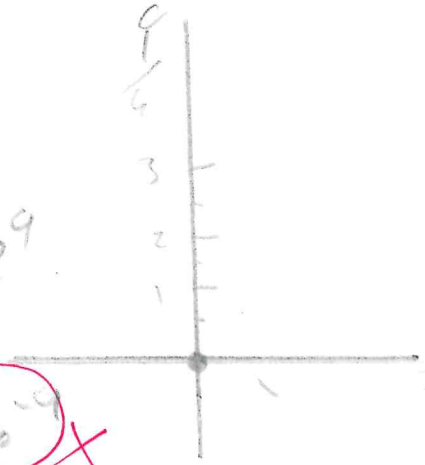
5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.



$$J = \begin{vmatrix} \frac{\partial x}{\partial v} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial y}{\partial u} \end{vmatrix} = \begin{vmatrix} e^u & ve^u \\ -e^{-u} & -ve^{-u} \end{vmatrix} = e^u + ve^{2u}$$

$$h_1 = h_v = (e^u)^2 + (e^{-u})^2 = \frac{e^{4u} + 1}{e^{2u}}$$

$$h_2 = h_u = (ve^u)^2 + (-ve^{-u})^2 = \frac{v^2 e^{4u} + 1}{e^{2u}}$$

$$dA = \left(\frac{e^{4u} + 1}{e^{2u}} \right) \left(\frac{v^2 e^{4u} + 1}{e^{2u}} \right) (du)(dv)$$

$$\int_0^3 \int_0^u \frac{1}{2} v^2 (e^{4u} + 1) dv du$$

$$= \frac{1}{2} \int_0^3 v^3 \big|_0^u du = \frac{1}{2} \int_0^3 u^4 du = \frac{1}{2} \cdot \frac{1}{5} u^5 \big|_0^3 = \frac{1}{10} (3^5) = 20.25$$

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Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const } (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

$$r^2 = x^2 + y^2$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$